

A Multi-Axis Robustness Analysis of Retrieval-Augmented Generation for Multi-Hop Requirements Traceability

Anonymous Author(s)

Anonymous Institution

Abstract. Reported verdicts on GraphRAG versus vector RAG disagree, and the evidence is typically tied to a single corpus, embedder, and judge – and, we show, to where citation quality is measured. We vary all four: a five-pipeline architecture matrix held fixed while we swap the embedder (e5-small 384d, text-embedding-3-small 1536d, bge-m3 1024d), the corpus (DO-178C typed-edge requirements, and Wikipedia paragraph chains via MuSiQue and 2WikiMultihopQA), the graph traversal (bounded 2-hop walk vs. personalized PageRank), and the judge (GPT-5.4, GPT-4.1 and Gemini 3.7-flash), over 13,400 generation runs at up to three seeds and more than 30,000 faithfulness judgments. Our central finding concerns the measurement point rather than the architecture: GraphRAG’s graph walk floods the context window at precision 0.12–0.23, while the synthesizer it feeds cites selectively at 0.48–0.98. Scoring the retrieved set as the attribution set therefore ranks GraphRAG below the vanilla baseline, while scoring the answer’s citations ranks it first of all pipelines – in all six settings we tested; the split survives a change of traversal and defeats two obvious mitigations, since the synthesizer already performs the filtering. Answer-level winners are stratum-conditional but stable across embedders and seeds. Faithfulness declines with hop distance on all three corpora once the comparison is adequately powered – a corpus-conditional reading that smaller judged subsets suggest does not survive – and the decline tracks context expansion rather than the corpus: the one pipeline that expands nothing stays flat, and a hop×expansion interaction is significant at $p=10^{-7}$. Judges, by contrast, are not stable: re-judging frozen answer–context pairs eleven weeks later gives self-agreement $\kappa \leq 0.14$; two vendors judging identical rows reverse which of them is stricter between corpora; and across eight settings their κ swings from 0.03 to 0.44 while raw agreement moves only from 0.82 to 0.94. We argue that RAG architecture claims should be tested at this level of robustness – including robustness to the citation-measurement point – before they are trusted.

Keywords: Retrieval-augmented generation · GraphRAG · Requirements traceability · Multi-hop QA · LLM-as-judge · Evaluation methodology

1 Introduction

Certifying avionics software under DO-178C requires demonstrating *traceability*: each requirement must be linked to the parent it derives from, the system-level intent it satisfies, and the verification artifact that tests it, and certification authorities increasingly require these chains to be auditable across two or more hops [24,26,7,8]. Requirements management tools record those links explicitly, so a requirements corpus is naturally read as a typed link graph, which makes retrieval over it a graph problem as much as a text problem.

RAG architectures for this task have produced contradictory verdicts: Microsoft GraphRAG [6] exploits structure but loses to vector RAG on detailed queries [15,31]; Adaptive-RAG [17] routes by complexity yet is hop-blind; LiSSA [10] and TVR [23] omit typed-graph reasoning. Prior evaluations report which architecture wins in a given setting, but not why the outcome changes when the setting changes.

We address this with a **multi-axis robustness** analysis. Holding a five-pipeline architecture matrix fixed (vanilla, agentic, agentic+graph, GraphRAG, learned adaptive), we vary four orthogonal axes: the retrieval embedder (e5-small 384d, Azure 3-small 1536d, bge-m3 1024d – three vendors, three dimensionalities), the corpus (DO-178C typed-edge requirements; Wikipedia paragraph chains via MuSiQue and 2WikiMultihopQA), the graph traversal (a bounded 2-hop walk and personalized PageRank over the same graph), and the faithfulness judge (GPT-5.4, GPT-4.1, Gemini 3.7-flash). The design produces 13,456 generation runs and over 30,000 faithfulness judgments.

Our headline result is not that one architecture wins, but that a methodological choice most evaluations leave implicit – whether citation quality is scored over the retrieved context or over the answer’s own citations – decides the winner. We make five contributions. **(C1)** Answer-level citation winners are *stratum-conditional but embedder- and seed-robust* (Table 2): vanilla ties GraphRAG on 1–2-hop DO-178C queries and is last at 3+–hop under all three embedders, where a graph-aided pipeline leads. **(C2)** GraphRAG’s walk and its synthesizer pull in opposite directions: the walk fills the context at precision 0.12–0.23, the answer cites 2.5–5.0 IDs at 0.48–0.98 (Table 3). Scoring the retrieved set as the attribution set – a choice some comparisons make implicitly – ranks GraphRAG below the vanilla baseline; scoring the answer’s citations ranks it first of all pipelines. This holds in all six settings, survives swapping the traversal, and defeats two mitigations. **(C3)** The faithfulness decline follows *context expansion*, not the corpus: powered comparably, both Wikipedia corpora decline too, overturning the corpus-conditional reading that smaller judged subsets suggest. The pipeline that expands nothing stays flat on every corpus, and a hop×expansion interaction is significant on the pooled matrix ($b=-0.599$, $p=1.0\times 10^{-7}$). **(C4)** *Judges are unstable along separable axes*: on frozen answer–context pairs, self-agreement falls to $\kappa\leq 0.14$ after eleven weeks; across eight settings two vendors reverse which of them is stricter, and their κ swings from 0.03 to 0.44 while raw agreement moves only from 0.82 to 0.94 (Table 4). Strictness is a judge–corpus property; the hop-wise ordering replicates across vendors regardless. **(C5, supporting)** a

logistic-regression router on dense embeddings alone reaches macro- F_1 0.86 on hop classification.

2 Related Work

Agentic and Adaptive RAG. Self-RAG [1] and CRAG [32] introduced reflective and corrective retrieval. Adaptive-RAG [17] routes by predicted query complexity; Probing-RAG [2] extends with internal-state probes; Search-R1 [18] trains agentic retrieval via RL; RAG-Critic [4] adds an iterative critic. None condition routing on typed-graph hop distance or expose a typed graph-lookup tool.

GraphRAG and multi-hop QA. Microsoft GraphRAG [6] and follow-ups [12,13,21] construct entity-relation graphs at indexing time. Han et al. [15] and GraphRAG-Bench [31] report that GraphRAG does not dominate vector RAG across all query types. Our C2 shows one reason such verdicts disagree: whether citation quality is scored over the retrieved set or over the answer’s citations decides which side wins, and C3 separates the context-flooding mechanism from its faithfulness consequence across corpora. Adjacent multi-hop benchmarks include MuSiQue [28], MultiHop-RAG [27] and 2WikiMultihopQA [16]; we use the first and the last to replicate C1, C2 and C3 outside the requirements domain.

RAG for requirements traceability. LiSSA [10], TVR [23], and Graph-RAG for compliance [22] evaluate single regulated domains without hop stratification or dual-judge protocols.

Citation evaluation, judge bias, and the κ paradox. ALCE [11] defined citation $P/R/F_1$ over generated answers; Wallat et al. [29] show that ALCE-style correctness is not faithfulness, motivating LLM judges. RAGChecker [25] provides a single-judge framework. Self-preference bias [30] motivates judge ensembles, and the κ paradox of Feinstein and Cicchetti [9], with Gwet’s AC1 [14], establishes the prevalence correction we apply. Our C4 adds a statistic this literature does not track: the judge’s agreement with itself under retrieval-state and date changes.

3 Method

3.1 Pipelines

All pipelines share embedder, ChromaDB vector store, a file-backed typed-edge graph, Azure GPT-5.4 generator, and grounded synthesis prompt; they differ only in retrieval. **(i) Vanilla:** dense top-10 retrieval, top-5 context, one synthesis call. **(ii) Agentic:** LangGraph [20] router-retriever-critic loop, `search_documents` tool only, iter cap 3. **(iii) GraphRAG:** vector seeds (8) + up-to-2-hop typed-graph walk (walk cap 30, context cap 15), single synthesis. This is a typed-edge local-walk retriever in the GraphRAG family, not Edge et al.’s community-summarization system [6]; we use “GraphRAG” as the family label. **(iv) Agentic+Graph:** (ii) plus a typed-edge `graph_lookup` tool. **(v) Adaptive:** chooses

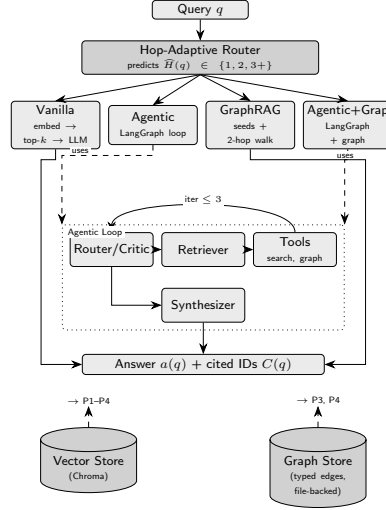


Fig. 1. Five-pipeline architecture with shared embedder, vector store, and typed-edge graph. Vanilla and GraphRAG bypass the agentic loop; Agentic and Agentic+Graph route through router/critic–retriever–tools ($\text{iter} \leq 3$) before synthesis. The adaptive pipeline (V1 rule-based or V2 learned) selects one of the four base pipelines per query.

among (i)–(iv) per query; we evaluate a rule-based router (V1) and a learned out-of-fold logistic regression (V2).

3.2 Robustness Axes

Embedder: three models from three vendors, at three dimensionalities, so the axis is not a family variant: e5-small (384d), Azure 3-small (1536d) and BAAI bge-m3 (1024d). ChromaDB collections are rebuilt per embedder; the graph is shared. **Traversal:** the production bounded undirected walk (≤ 2 hops, cap 30) and, as a second retriever of the same family, personalized PageRank with restart from the vector seeds – unbounded in radius and ranked by random-walk mass, with embedder, seed count and context cap held fixed so the traversal is the only difference. **Corpus:** a synthetic DO-178C-style aerospace certification corpus of 1,132 requirements over 30 modules (17–76 requirements each; median 59, p90 103 tokens per requirement), generated by GPT-5.4 from a hand-authored module taxonomy and publicly released with its link annotations; and two Wikipedia paragraph-chain sets: a 200-query MuSiQue [28] subset (67/67/66 across 2/3/4-hop, mapped to our 1/2/3+-hop strata, *REFERENCES* edges only between consecutive supporting paragraphs) and a 200-query 2WikiMultihopQA [16] subset. 2Wiki carries two or four supporting titles and never three, so its strata come from reasoning structure rather than paragraph count: *comparison* (two independent lookups) is 1-hop, *compositional* and *inference* (chained, the second needs the first’s answer) are 2-hop, and *bridge_comparison* (four evidences) is 3+-hop.

Retrieval traverses the requirements graph restricted to five traceability edge types (*derives_from*, *satisfies*, *references*, *verifies*, *refines*); system-engineering edges such as allocation are excluded as not retrieval-relevant. **Judge:** GPT-5.4, GPT-4.1 and Gemini 3.7-flash. The full matrix is judged in its entirety rather than in 300-row subsets, so per-stratum n is ≈ 300 rather than ≈ 33 , and every cross-corpus or cross-vendor comparison is drawn from a single judging window.

3.3 Queries and Gold Evidence

DO-178C queries are built from the graph, not from free-form prompting: we sample directed requirement–requirement chains of length 1, 2, and 3–4 over the five whitelisted edge types and prompt GPT-5.4 to phrase one natural-language question answerable from the chain. The chain’s node set is the gold evidence, so gold size grows with the stratum (mean 2.00/3.00/4.68 IDs for 1/2/3+-hop). This yields 296 queries (100/99/97 per stratum), 85% of them cross-module, spanning 30 source and 26 target modules; chain edges are dominated by *references* (497 of 655), then *satisfies* (114), *verifies* (36), *refines* (5) and *derives_from* (3). Gold sets are therefore structural rather than human-annotated: a sampled chain is by construction a supporting path, but we do not verify that no *other* path also answers the question, so citation recall is exact while precision is conservative wherever alternative evidence exists. This affects all pipelines identically and so does not bias the comparisons.

3.4 Metrics and Judging

We report ALCE-style citation $P/R/F_1$ [11] against gold ID sets. Cited IDs are parsed from the answer text with a vocabulary-anchored parser (exact corpus IDs, longest match first); hallucinated IDs that reuse a real module prefix count against precision, and incidental tokens (SHA-256, DO-254) are excluded. We separately report *context precision* – the gold fraction of the retrieved set handed to the synthesizer – and retrieval recall, since conflating the context set with the answer’s citations changes which architecture appears to win (§5.2). Each faithfulness judgment is a strict-JSON binary verdict over the retrieved-context block. We report Cohen’s κ , Gwet’s AC1 [14,9], raw agreement, and McNemar exact-binomial p ; per faithfulness cell, Wilson 95% CIs and Cochran–Armitage hop-trend tests; and three same-judge controls (test–retest, embedder swap, eleven-week re-judge) on a paired 300-tuple subset (§5.4).

3.5 V2 Learned Router

14 features per query: 11 hand text features (ID regex, keyword flags, log token length) and 3 PCs of the query embedding (fit per fold). Multinomial logistic regression, L_2 ($C=1$), Platt-calibrated, under stratified 10-fold $\times$ 5-repeat CV (50 fits). The per-stratum routing target is the empirical mean- F_1 winner, derived once from the full locked matrix (not re-fit per fold); only hop prediction is out-of-fold, reported for all 296 queries.

4 Experimental Setup

Main matrix and cross-corpus sample. DO-178C: 5 pipelines $\times$ 296 hop-stratified queries $\times$ 3 seeds = 4,440 runs under each of two embedders (*v2*, *v3*), plus 1,480 runs at one seed under the third (*v4*, bge-m3) and a 296-run personalized-PageRank arm on the Azure embedder. MuSiQue: 3 pipelines (vanilla, GraphRAG, agentic+graph) $\times$ 200 queries $\times$ 1 seed = 600 runs, Azure embedder only, with the GraphRAG arm extended to three seeds so its per-stratum n matches DO-178C’s, plus 400 runs for two mitigation variants and a 200-run GraphRAG distractor-edge control (§5.1) on the same queries and vector seeds. Agentic and adaptive are not re-run on MuSiQue (C2 needs only the three-pipeline trio). 2WikiMultihopQA: vanilla and GraphRAG $\times$ 200 queries $\times$ 3 seeds = 1,200 runs, Azure embedder.

Faithfulness judging. On DO-178C, the multi-judge protocol (GPT-5.4 + GPT-4.1) is applied to two disjoint 300-row stratified batches (60 per pipeline; seeds 42 and 43), the second judged eleven weeks after the first, yielding 1,200 paired binary judgments per embedder. Each batch is *pinned* to the same 300 (query, pipeline, repeat) tuples across embedders so C3 deltas are computed on identical items; batches are reported separately, never pooled across judging dates. On the Wikipedia corpora we judge every row with GPT-4.1 and Gemini 3.7-flash. The Gemini leg of the MuSiQue seed extension, both mitigation arms and 2Wiki ran one day after their GPT-4.1 leg, on frozen stored answer-context strings, so the pairing is row-exact; the DO-178C matrices were judged by both vendors in a single pass. Same-input controls re-judge 300 v2 tuples twice with GPT-5.4 (test-retest) and once eleven weeks later (temporal drift); a generator-swap control re-synthesizes 332 v3 rows with GPT-4.1 on frozen retrievals and dual-judges them.

Statistical protocol. 95% BCa bootstrap intervals [5] ($B=1000$, paired at the query level), Wilcoxon signed-rank with Holm correction across the 30 pipeline-pair $\times$ stratum contrasts [3,19], and Cliff’s δ (negligible <0.147). A pipeline pair is reported as significantly different only when the BCa interval excludes zero, Holm $p < 0.05$, and $|\delta| \geq 0.147$ jointly hold. Generation is stochastic, so every query is run at three seeds; tests pool the seeds and we report the across-seed spread separately (§5.1) as a check that no reported ordering rests on a single run.

Reproducibility. The graph store was populated before this study and carries edges beyond those annotated in the released corpus: re-loading the release with our ETL recovers 91.0% of the edges and 82.8% of the chains behind our queries. Graph-pipeline results are therefore reproducible up to that coverage; the query manifest is released with its full chains, so every gold evidence set is exact regardless. The rest of the pipeline regenerates from a single make target on locked CSVs. Azure model snapshots: `gpt-5.4-2026-03-05` (generator + judge), `gpt-4.1-2025-04-14` (judge); reasoning-mode decoding with an 8,192-token completion cap. The MuSiQue subgraph (3,996 paragraph chunks, 399 REFERENCES edges) is built deterministically from the `dgslibisey` MuSiQue mirror (validation split, seed 20260511), and the 2WikiMultihopQA subgraph

Table 1. Per-stratum citation F_1 across four settings. v2 main = DO-178C synthetic with local e5-small embeddings (4,440 runs); v3 main = same corpus with Azure text-embedding-3-small (4,440 runs); MuSiQue = 200-query Wikipedia stratified subset with Azure embedder (600 runs, vanilla / agentic-graph / graphrag only); 2Wiki = 200-query 2WikiMultihopQA subset at three seeds (1,200 runs, vanilla / graphrag only). Bold marks the per-(setting, stratum) maximum. The stratum-conditional pattern is the same on all four: vanilla is competitive at 1-hop and loses as hop distance grows.

System	v2 main (local)			v3 main (Azure)			MuSiQue (Azure)			2Wiki (Azure)		
	1h	2h	3+h	1h	2h	3+h	1h	2h	3+h	1h	2h	3+h
vanilla	0.757	0.712	0.089	0.756	0.689	0.195	0.742	0.415	0.277	0.990	0.778	0.678
agentic	0.541	0.414	0.175	0.533	0.430	0.203	—	—	—	—	—	—
agentic-graph	0.566	0.415	0.219	0.547	0.421	0.251	0.457	0.296	0.249	—	—	—
graphrag	0.751	0.720	0.172	0.745	0.707	0.253	0.841	0.632	0.463	0.969	0.946	0.938
adaptive	0.630	0.499	0.181	0.651	0.492	0.254	—	—	—	—	—	—

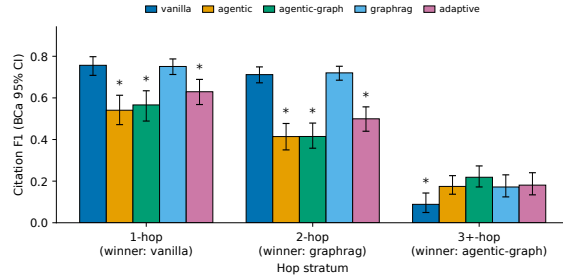


Fig. 2. Per-stratum citation F_1 on the v2 main matrix (DO-178C, local e5-small embedder): five pipelines with BCa 95% CIs; * marks pipelines significantly different from the per-stratum winner (Holm-Wilcoxon $p < 0.05$, $|\delta| \geq 0.147$).

(2,000 chunks, 332 edges) from the `voidful` mirror (seed 20260816) by the same procedure. The v2/v3 graph arms walked that original store; the bge-m3 matrix and the PPR arm walked a file-backed store built from the released link annotations, which is also what a reproduction uses (no database service needed). To check that this change of graph does not drive the comparisons, we rebuilt every v3 GraphRAG context on the file-backed store without any generation: the vector seeds are identical in 295 of 296 queries, the contexts overlap at Jaccard 0.89, and context precision (0.131 vs. 0.129) and retrieval recall (0.720 vs. 0.724) differ by at most 0.01 in every stratum. The citation side was not re-generated on the new store.

5 Results

5.1 C1: Winners are Corpus-Conditional, Embedder- and Seed-Robust

Table 1 reports per-stratum F_1 across four settings. On DO-178C the pattern repeats under both embedders: vanilla and GraphRAG are statistically tied on 1-hop and 2-hop (Wilcoxon $p=0.12/0.65$ local, negligible δ), the agentic loop costs 0.19–0.30 F_1 on those strata, and on 3+-hop the ordering inverts – vanilla drops to last and the graph-aided pipelines lead (local embedder: agentic-graph 0.219 vs. vanilla 0.089, Holm $p<10^{-6}$; under the Azure embedder the same ordering holds but no 3+-hop pair passes the joint significance criterion). A third embedder from a third vendor, BAAI **bge-m3** (1024d), reproduces it again (Table 2): vanilla is last at 3+-hop under all three (0.089 / 0.195 / 0.114) while a graph-aided pipeline leads under all three, and under bge-m3 both graph arms beat vanilla at 3+-hop (+0.100 and +0.091, Holm $p=2.1\times 10^{-4}$ and 1.4×10^{-3}) while the 1-hop and 2-hop contrasts stay non-significant with negligible δ . On MuSiQue (three-pipeline field) GraphRAG wins every stratum outright ($F_1=0.841/0.632/0.463$; vs. vanilla, Wilcoxon $p\leq 5.1\times 10^{-4}$, $\delta=0.27\text{--}0.43$). Because the MuSiQue graph’s *REFERENCES* edges connect only gold supporting paragraphs, we re-ran the GraphRAG arm on a control graph with consecutive-distractor edges added (context precision falls 0.23→0.14; the walk cap is always reached). GraphRAG loses only 0.02/0.05/0.09 F_1 per stratum and still beats vanilla everywhere (+0.08/ +0.16/ +0.10, $p\leq 0.014$): the MuSiQue win is not an artifact of gold-only edges. A second Wikipedia corpus repeats the stratum-conditional shape a third time: on 2WikiMultihopQA at three seeds, vanilla and GraphRAG are tied at 1-hop (0.990 vs. 0.969, $p=0.15$, $\delta=-0.015$) and GraphRAG wins 2-hop and 3+-hop by widening margins (+0.168, $\delta=0.48$; +0.260, $\delta=0.87$; both $p<10^{-19}$) – the same crossover as DO-178C and MuSiQue, now under a corpus whose graph we did not design. Seed variance is small against these effects: across the three seeds, per-pipeline mean F_1 moves by at most 0.031 (max SD 0.016), and every ordering we report as significant holds independently in all three seeds; the only rank swaps are between vanilla and GraphRAG on 1–2-hop, the pair we report as tied. Which architecture wins is thus corpus- and stratum-conditional, but stable under embedder swap and reseeded.

5.2 C2: The Walk Floods the Context, the Synthesizer Filters It

Table 3 reports the same measurement in six settings: three embedders, three corpora, and two traversals. GraphRAG’s walk fills 11–15 context slots at context precision 0.12–0.23 – four to five of every six retrieved chunks are off-gold. The synthesizer then cites only 2.5–5.0 IDs per answer, at citation precision 0.48–0.98: a 2.9–4.2× precision enrichment over its own context, in every setting. GraphRAG’s F_1 edge is therefore a *recall* effect: the walk surfaces gold chunks dense retrieval misses, and the synthesizer declines to cite most of the noise that rides along. Evaluations that score the retrieved set as the attribution set

Table 2. Per-stratum citation F_1 on DO-178C under three independent embedders (384d Microsoft, 1536d OpenAI, 1024d BAAI). The stratum-conditional ordering is stable: vanilla ties GraphRAG on 1–2-hop and is *last* at 3+-hop in all three, where a graph-aided pipeline leads.

System	e5-small (384d)			3-small (1536d)			bge-m3 (1024d)		
	1h	2h	3+h	1h	2h	3+h	1h	2h	3+h
vanilla	.757	.712	.089	.756	.689	.195	.769	.691	.114
agentic	.541	.414	.175	.533	.430	.203	.548	.404	.156
agentic-graph	.566	.415	.219	.547	.421	.251	.546	.431	.206
graphrag	.751	.720	.172	.745	.707	.253	.742	.728	.214
adaptive	.630	.499	.181	.651	.492	.254	.672	.478	.202

Table 3. GraphRAG floods the context and the synthesizer filters it (C2), across every setting we ran: three embedders, three corpora, and two traversals. The walk fills 11–15 slots at low precision; the answer cites 2.5–5.0 IDs at 3–4 $\times$ that precision. The ranking inverts with the measurement point in all six.

Setting	<i>retrieved context</i>		<i>answer citations</i>		
	slots	prec.	IDs	prec.	gain
DO-178C, e5-small	14.9	0.120	4.9	0.480	4.0 $\times$
DO-178C, 3-small	14.9	0.129	5.0	0.493	3.8 $\times$
DO-178C, bge-m3	14.8	0.121	4.7	0.490	4.1 $\times$
DO-178C, PPR walk	14.9	0.131	4.9	0.509	3.9 $\times$
MuSiQue	11.2	0.227	3.4	0.654	2.9 $\times$
2WikiMultihopQA	11.4	0.230	2.5	0.975	4.2 $\times$

– as some GraphRAG comparisons do – measure the flooding and miss the filtering, and invert the ranking: by context precision GraphRAG sits below the vanilla baseline in every setting (0.12–0.23 against 0.27–0.37), while by answer-citation F_1 it is the highest-scoring pipeline in every setting (0.551, 0.571, 0.565, 0.585, 0.646, 0.951 in the row order of Table 3). The two measurements do not merely differ in magnitude; they disagree on the sign of the comparison that the architecture debate turns on.

Two controls show the split is not an artifact of our own design choices. First, *traversal*: replacing the bounded 2-hop walk with personalized PageRank over the same graph – unbounded in radius and ranked by random-walk mass rather than hop distance, with embedder, seeds and context cap held fixed – changes almost nothing (context precision 0.131 vs. 0.129, and 0.131 for the 2-hop walk rebuilt on the same file-backed graph, §4; enrichment 3.9 $\times$ vs. 3.8 $\times$; citation F_1 +0.016 at Cliff’s $\delta=0.036$, which our joint criterion rejects as negligible). Second, *mitigation*: neither obvious fix helps. Reranking the walked neighbours by query similarity and keeping the top five leaves everything where it was (context precision 0.227 $\rightarrow$ 0.232, citation F_1 +0.010, $p=0.34$), while tightening the

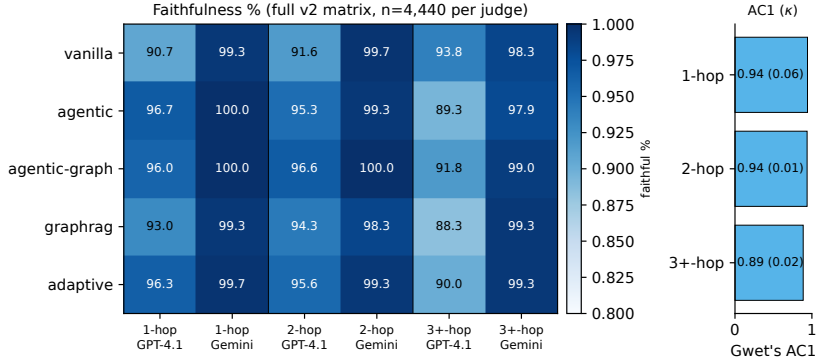


Fig. 3. Faithfulness on the full v2 main matrix (DO-178C, e5-small; 4,440 rows, ≈ 300 per cell), judged by GPT-4.1 and Gemini 3.7-flash on identical rows. *Left:* under GPT-4.1 every expanding pipeline loses 4–7 pp from 1-hop to 3+-hop while vanilla, which expands nothing, does not decline (C3); Gemini compresses every cell against the ceiling. *Right:* per-stratum Gwet’s AC1 with Cohen’s κ in parentheses – 0.89–0.94 against $\kappa \leq 0.06$ on the same verdicts, the prevalence paradox of C4. Colour scale starts at 80%.

traversal budget actively hurts (walk $30 \rightarrow 10$, context $15 \rightarrow 8$: citation $F_1 -0.126$, $p=2.6 \times 10^{-11}$, and context precision *falls* to 0.194). This is what C2 predicts: the synthesizer already performs the filtering, so filtering again upstream is redundant, and cutting the budget only removes the recall the advantage rests on.

5.3 C3: The Decline Follows Context Expansion, Not the Corpus

Judged on 300-row subsets, the faithfulness decline looks *corpus-conditional*: present on DO-178C, absent on Wikipedia chains. Judging the full matrix rather than 300-row subsets, bringing the MuSiQue arm up to comparable power, and adding a second Wikipedia corpus overturns that reading.

The MuSiQue arm had 67 rows per stratum, since 200 queries split three ways. Its observed 1-hop to 3+-hop drop was 9.2 pp – larger than the 4.7 pp drop that reaches significance on DO-178C – but detecting it at 80% power needs about 210 rows per cell. Re-running that arm at three seeds ($n=600$, ≈ 200 per cell) makes the decline significant under both judges: 0.930/0.876/0.823 under GPT-4.1 ($p=1.1 \times 10^{-3}$) and 0.955/0.945/0.869 under Gemini ($p=1.2 \times 10^{-3}$). 2WikiMultihopQA, a second Wikipedia corpus judged in full ($n=1,200$, both judges), declines as well ($p=5.0 \times 10^{-2}$ and 4.8×10^{-6}). The apparent corpus-conditionality was an artifact of an underpowered cell, and the decline now replicates on three corpora under two vendors.

What does separate pipelines is whether they expand the context at all. Vanilla is the only pipeline that retrieves a fixed top-5 and expands nothing, and its 1-hop to 3+-hop drop is statistically indistinguishable from zero on both

corpus families under GPT-4.1: -1.5 pp on DO-178C ($n=2,072$, per-embedder $p=0.16, 0.99, 0.76$) and $+1.1$ pp on 2Wiki ($n=600, p=0.66$). The four expanding pipelines drop 6.3 pp and 6.7 pp on the same two corpora. Fitting faithfulness on hop index, an expansion indicator and their interaction makes the contrast a single test: the interaction is negative and significant on the pooled DO-178C matrix ($b=-0.599, p=1.0\times 10^{-7}, n=10,360$) and negative with the same magnitude on 2Wiki ($b=-0.286, p=0.27, n=1,200$, where the arm is a twentieth the size). Expansion does not merely accompany the decline; it steepens it.

Under Gemini the same ordering holds but the test does not resolve: at 0.97 – 1.00 faithful across most DO-178C cells the matrix is compressed against the ceiling, the baseline itself acquires a slope ($b=-0.729, p=0.012$), and the interaction is no longer separable from it ($p=0.11$). The expanding arms still drop further than vanilla in both corpora (2.8 vs. 1.5 pp on DO-178C, 12.7 vs. 7.7 pp on 2Wiki). This is C4 acting on C3: a lenient judge preserves the ranking and loses the resolution.

We therefore state what the data supports: the non-expanding baseline is flat, expansion steepens the hop-wise decline, and the effect is measurable wherever the judge is not saturated. We do not claim every expanding pipeline declines in every cell – GraphRAG is marginal under bge-m3 ($p=0.065$), and the agentic arms decline in four of six embedder–pipeline combinations. The GPT-5.4 leg, on which the subset-level corpus-conditional reading also rested, has not been re-run at this power.

5.4 C4: Judges Are Unstable on Frozen Inputs and on Changed Ones

Table 4 separates judge instability into a component measured on *frozen* inputs, where no retrieval is re-run, and one measured under a changed retrieval state. *Frozen inputs*. Re-judging the identical 300 (answer, context) pairs – the stored strings, byte for byte – gives GPT-5.4 $\kappa=0.76$ (raw agreement 0.88) on the same day, but only $\kappa=0.14$ (0.56) eleven weeks later, a $+15$ pp leniency shift. Sampling noise cannot explain the gap: under a stationary judge, cross-date agreement should sit at the same-day level, and the observed 0.56 rejects stationarity (binomial $p<10^{-44}$). GPT-4.1 on the same frozen tuples is starker: $\kappa=-0.05/-0.00$ (v2/v3), chance-level agreement with its own earlier verdicts, a $+41$ pp leniency shift. With the inputs held fixed, this component is judge drift alone. *Changed retrieval state*. Re-judging after an embedder swap – which changes retrieved contexts and answers on 94% of tuples – gives GPT-5.4 $\kappa=0.137$ (41% of verdicts flip) against the 0.76 same-day floor; GPT-4.1 flips only 26% ($\kappa=0.48$). This component mixes output drift with judge response, and the two judges rank oppositely on the two axes: the judge more stable under embedder swap is the less stable across time, so stability on one axis does not transfer to the other. *Across vendors*. Adding a third vendor (Gemini 3.7-flash, thinking enabled) as a judge on identical rows sharpens the picture, and shows that leniency is not a fixed property of a judge. On the four DO-178C settings Gemini is the more

Table 4. Judge stability. *Top*: same judge, frozen input — the answer and context strings are identical and only the judging date differs, so the fall from 0.76 to 0.14 isolates drift from any retrieval effect. *Middle*: same judge, input changed by embedder swap. *Bottom*: cross-vendor agreement on identical rows. Which judge is stricter depends on the corpus — Gemini is markedly more lenient on DO-178C, indistinguishable on the capped MuSiQue arm, and stricter on 2Wiki. Raw agreement and AC1 move together across the eight settings while κ swings by a factor of fifteen, ordered by distance from the ceiling.

<i>Same judge, frozen input (answer + context identical)</i>				
GPT-5.4, re-judged same day				$\kappa = 0.764$, raw agr. 0.88
GPT-5.4, re-judged 11 wk later				$\kappa = 0.138$, raw agr. 0.56
GPT-4.1, re-judged 11 wk later	$\kappa = -0.05 / -0.00$	(v2/v3), raw agr. 0.48 / 0.51		
GPT-4.1, full matrix vs. May				$\kappa = -0.046$, raw agr. 0.48, +43 pp
<i>Same judge, input changed by embedder swap</i>				
GPT-5.4 (v2 vs v3)				$\kappa = 0.137$, raw agr. 0.59
GPT-4.1 (v2 vs v3)				$\kappa = 0.480$, raw agr. 0.74
<i>Cross-vendor, same rows, one judging window (Gemini 3.7-flash $\times$ GPT-4.1)</i>				
<i>Setting</i>	raw	κ	AC1	faithful (G / 4.1)
DO-178C e5-small ($n=4,440$)	0.929	0.029	0.923	0.993 / 0.933
DO-178C 3-small ($n=4,440$)	0.913	0.125	0.903	0.980 / 0.917
DO-178C bge-m3 ($n=1,480$)	0.939	0.047	0.935	0.991 / 0.945
DO-178C PPR walk ($n=296$)	0.926	0.058	0.919	0.983 / 0.936
MuSiQue, 3 seeds ($n=600$)	0.850	0.172	0.817	0.923 / 0.877
MuSiQue rerank ($n=200$)	0.820	0.171	0.772	0.935 / 0.825
MuSiQue budget cap ($n=200$)	0.875	0.419	0.841	0.875 / 0.880
2WikiMultihopQA ($n=1,200$)	0.899	0.435	0.877	0.888 / 0.914

lenient of the pair by a wide margin – 0.98–0.99 faithful against GPT-4.1’s 0.92–0.95, with McNemar rejecting symmetric disagreement everywhere ($p < 10^{-9}$). On 2WikiMultihopQA the direction *reverses* (0.888 vs. 0.914, $p = 6.2 \times 10^{-3}$), and on the budget-capped MuSiQue arm the two are indistinguishable (0.875 vs. 0.880, $p = 1.00$). Which judge is stricter is therefore a property of the judge–corpus pair, not of the judge, and cannot be calibrated away once and reused.

The same eight settings make the prevalence paradox of Feinstein and Cicchetti [9] directly visible in a single judge pair. Raw agreement moves only from 0.82 to 0.94 across them, and Gwet’s AC1 tracks it closely (0.77–0.94), but Cohen’s κ swings from 0.03 to 0.44 – ordering the settings almost exactly by how far faithfulness sits from the ceiling (Table 4, bottom). At 93% agreement on DO-178C, reporting κ alone would license the claim that two judges agree no better than chance; the same pair on 2Wiki, agreeing *less* often (0.899), would appear fifteen times more reliable ($\kappa = 0.435$ against 0.029).

Crucially, leniency and *trend* come apart — despite sitting near the ceiling on DO-178C, Gemini still reproduces the hop-wise decline there ($p = 4.3 \times 10^{-3}$

Table 5. V2 router replication and Azure-embedder boost. Same 14-feature schema (11 hand text features, 3 PCs of the query embedding); only the embedder differs. Text features remain mostly inert (4/11 non-zero), so the +0.08 macro- F_1 gain is attributable to dense-embedding hop-decodability.

	v2 (local 384d $\rightarrow$ PCA-3)	v3 (Azure 1536d $\rightarrow$ PCA-3)
50-fold macro- F_1	0.78 ± 0.08	0.86 ± 0.05
Adaptive-V2 overall F_1	0.559	0.569
Adaptive-V1 overall F_1	0.439	0.468
Oracle overall F_1	0.641	0.652
V2 – V1 (absolute)	+0.120	+0.101
Gap closure (V1 $\rightarrow$ Oracle)	59.4%	54.8%
Holm-Wilcoxon strata sig	3/3	2/3

and $p < 10^{-16}$ on the two main matrices) and on both Wikipedia corpora. A judge’s strictness and the ordering it induces are separable properties: the second replicates across vendors, the first does not, though a saturated judge loses the power to resolve interactions (§5.3).

Between judges, and why we re-judged everything. Inter-judge κ halves under embedder swap on identical tuples ($0.30 \rightarrow 0.17$ overall); McNemar rejects symmetric disagreement (exact $p < 0.001$). On the 300-row MuSiQue batch the GPT-5.4/GPT-4.1 pair agreed at chance level ($\kappa = 0.02$, raw agreement 0.48, Gwet’s AC1 0.08 – genuine disagreement, not a prevalence artifact; GPT-4.1 judged 88% faithful against GPT-5.4’s 47%), and the replication batches show the same at scale: eleven weeks on, GPT-4.1 marks 93–95% of fresh same-distribution tuples faithful, inter-judge κ falls to 0.05/0.01 (local/Azure), and the hop trend GPT-4.1 itself had established in the original batches disappears. Verdicts of that instability cannot carry a cross-corpus claim, which is why the corpus-conditional reading built on those batches did not survive re-judging the full matrix inside a single window (§5.3). We report no faithfulness comparison that spans judging dates.

C5 (supporting): Dense embeddings alone classify hop. Table 5 shows the V2 router replication. The 50-fold macro- F_1 improves $0.78 \pm 0.08 \rightarrow 0.86 \pm 0.05$ under embedder swap, and Adaptive-V2 closes 59%/55% of the V1 $\rightarrow$ Oracle gap (local/Azure), with Holm-significant per-stratum gains in 3/3 strata locally and 2/3 under Azure (the 3+-hop routing targets are statistically tied there). Only 4 of the 11 hand text features are ever non-zero on the 296 queries; the rest are inert. The router therefore operates almost exclusively on the 3 query-embedding PCs, suggesting hop distance is largely an embedding-decodable property of the query text in our generated query set; transfer to human-authored queries is untested.

6 Discussion

Why does expansion cost faithfulness? GraphRAG’s traversal expands candidates well beyond what dense retrieval surfaces, and the synthesizer filters most of the noise out of its citations (C2). What it cannot filter is the influence of that noise on the answer itself: faithfulness is judged against the full retrieved block, so every chunk the walk adds is a chunk the answer can be found unfaithful to. That mechanism is indifferent to what the edges connect, which is why the decline appears on typed DO-178C traceability links and on Wikipedia paragraph adjacency alike, and why the one pipeline that adds nothing is the one that stays flat. The cost scales with the number of expanded chunks rather than with their semantic distance — consistent with C2’s finding that reranking those chunks for relevance changes nothing.

Implications for LLM-judge practice. Same-judge self- κ across embedders is 0.137, against a same-day test–retest floor of 0.76; on frozen inputs eleven weeks apart, self- κ is 0.14 (GPT-5.4) and -0.05 (GPT-4.1). A single-judge comparison across retrieval modules therefore reports embedder- and date-conditional verdicts, not invariant faithfulness. Multi-judge ensembles mitigate this only partially, since inter-judge κ itself halves under embedder swap. Paired-tuple judging under a fixed embedder is the least that is needed for comparable faithfulness numbers.

Threats to validity. Construct. ALCE-style F_1 does not measure rationale quality; we pair it with multi-judge faithfulness, while acknowledging LLM-judge bias [30,29]; C4 quantifies this directly. A third judge from a different vendor (Gemini 3.7-flash) now covers every matrix in the study, and C4 shows its strictness relative to GPT-4.1 reverses between corpora. *External.* The DO-178C-style corpus is a single synthetic aerospace dataset, generated by the same model family that answers and judges; absolute F_1 levels characterize this single regulated-domain setting and may not transfer. A generator-swap control (GPT-4.1 re-synthesizing 332 rows on frozen retrievals) reproduces the architecture ordering with slightly higher F_1 , and its answers show the same hop-wise faithfulness decline under both judges, so neither citation nor faithfulness results are GPT-5.4-specific; the corpus-author leg of the circularity remains. Two Wikipedia corpora (MuSiQue, 2WikiMultihopQA) are the main external-validity check for the universality claim; both are judged in full by two vendors and both replicate C1’s stratum-conditional crossover, though only DO-178C carries the full five-pipeline field. Our graph pipeline is one typed-edge walk implementation, though swapping it for personalized PageRank leaves C2 intact; a full third-party GraphRAG system may still flood or filter differently. *Mitigations we tried.* Two obvious fixes fail. Reranking the walked neighbours by query similarity and keeping the top five moves nothing (citation $F_1 +0.010$, $p=0.34$); tightening the traversal budget hurts and does not even buy precision (citation $F_1 -0.126$, $p=2.6\times 10^{-11}$, context precision $0.227\rightarrow 0.194$). Both are consistent with C2: the synthesizer is already the filter, so filtering upstream is redundant

while budget cuts remove the recall the advantage depends on. A learned evidence selector conditioned on the question, rather than the two unsupervised heuristics we tried, remains open. *Statistical*. Per-stratum n for citation metrics is 97–100 on DO-178C and 66–67 on the Wikipedia corpora (paired at the query level), the lower edge of BCa stability under heavy skew; a percentile fallback shows no qualitative reversal. Faithfulness cells now carry $n \approx 300$ after full-matrix judging, which is what overturned the corpus-conditional reading in C3. The hop $\times$ expansion interaction is significant on DO-178C but only same-signed on 2Wiki, whose arm is a twentieth the size; we report it as directional there. *Reranker / router engineering*. No cross-encoder reranker is included (orthogonal axis); C5 is a diagnostic instrument, not a production router.

Artifact availability. The AeroSys corpus (1,132 requirements, 30 modules, with its link annotations), the 296-query manifest with full chains, the pipeline code and every scored and judged CSV behind the tables are released under CC-BY-4.0 (links withheld for anonymous review). The Wikipedia subsets are rebuilt from public mirrors by seeded scripts in the same release.

7 Conclusion

Varying embedder, corpus, traversal and judge around a fixed five-pipeline matrix: *graph expansion floods the context but the synthesizer cites selectively, and moving the measurement point from context to answer inverts the architecture ranking* – in all six settings, under a changed traversal, and against two mitigations; *the faithfulness decline follows context expansion rather than the corpus*, replicating on three corpora under two vendors overturning the corpus-conditional reading that smaller subsets suggest; *stratum winners are embedder-and seed-robust*; and *judges drift on frozen inputs, reverse which of them is stricter between corpora, and do so while κ hides it*. Robustness on these axes – and an explicit choice of citation-measurement point – is the minimum bar for RAG architecture claims.

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